

Ángel Alberto Carretero

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EXPERIENCE

- 09/2025–present **Performance and Correctness – Office of the CTO, CANONICAL (REMOTE)**
My current focus is on correctness of distributed systems using formal (TLA+/TLAPS) and semi-formal methods (state machines, property-based testing, model-based testing). Some highlights:
- Implemented Raft in Rust in a very short timeframe through careful use of TLA+ for model-based testing of the code and discrete event simulation.
 - Designed the core protocols using Hierarchical State Machines (HSM) which allows us to reason about correctness upfront and produced code that was significantly easier to read and to debug.
 - Led some company-wide sessions on TLA+ with a focus on producing careful designs which substantially decrease the cost of implementing distributed systems.
 - Co-designed a performant storage layer based on SQLite which uses a custom VFS.
- Main skills:** Rust, Tokio, distributed systems, systems programming, SQLite, TLA+/TLAPS, state machines, property-based testing, discrete event simulation.
- 09/2023–09/2025 **Software Engineer – Office of the CTO, CANONICAL (REMOTE)**
Worked directly with the CTO to kickstart challenging projects used across Canonical. Some highlights:
- Lead developer for [chisel](#), a widely-used tool written in Go to create containers with small storage footprint and a reduced attack surface.
 - Contributor to [dqlite](#), a distributed database using Raft and SQLite, written in C. I developed a snapshot protocol to bring followers up to date, and I used state machines to reason about termination.
 - As part of my duties, I supervised and mentored developers, led planning, estimating, and architecting. In a few months I managed to deliver 1.0 for a project that had been stagnant for a year.
- Main skills:** Deep experience in Go and its runtime, containers and OCI images, GNU/Linux.
- 09/2022–09/2023 **Software Developer Engineer I – Books Detail Page, AMAZON (MADRID)**
My team owned the book detail pages in all surfaces across [www.amazon.com](#), a Tier-1 service that receives thousands of requests per second. Some highlights:
- Developed features using Java/Spring for the backend, JSPs and JS for the frontend. This required complex debugging across many different architectures and surfaces on a custom stack with dozens of layers and many different teams working on the same codebase simultaneously.
 - Led a successful project coordinating two different teams in different organizations, PMs and design. Among my different duties, I carefully designed the experiment to ensure precise measurement of KPIs.
 - Debugged production issues across a multi-layer Tier-1 service and participated in on-call for Amazon's primary website.
- Main skills:** Full stack, driving features from experiment planning to production deployment.

- 09/2021–06/2022 **Researcher (Collaboration grant) – Computer Science department, UAM (MADRID)**
Designed an esoteric programming language and created an interpreter from scratch in Rust which included a library of parser combinators, a tree-walk interpreter, a typechecker, and more.
Main skills: Rust, programming language theory.
- 06/2021–09/2021 **Software Developer Engineer Intern – Music ML, AMAZON (BERLIN)**
Worked on recommendation systems for Amazon Music which used an orchestrator written in functional async Java.
Main skills: Async functional Java, experiment design.

EDUCATION

- 2017–2022 **BSc in Mathematics, UNIVERSIDAD AUTÓNOMA DE MADRID**
- 2017–2022 **BSc in Computer Science, UNIVERSIDAD AUTÓNOMA DE MADRID**
- Combined average of 9.02/10, 14 courses with honors (top 2%). Awarded three times the annual scholarship for the top 1% of all university students in Madrid.
 - Mathematics Bachelor's thesis: *Pseudo-Riemannian Geometry and Schwarzschild spacetime*. (9.9/10)
 - Computer Science Bachelor's thesis: *Formal design and implementation of a programming language based on facets*. (9.5/10)

TALKS

Building a Reliable Database Protocol with AI, Formal Methods, and a Simulation Framework – BUG BASH - GOTO 2026

Git gud: how Git works internally – OPEN SOUTH CODE 2024
Explore how Git stores its state on disk inside the .git subdirectory. For each command, inspect the results live using low-level tools to gain a deeper intuition.

LANGUAGES

Spanish First Language.

English C1 Level, certified Cambridge English Advanced.

CONTACT

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